

# MICHAEL TOMMASO

(917) 679-3307 | Mjt288@cornell.edu | linkedin.com/in/MichaelTommaso | michaeltommaso.com | github.com/michaeljtommaso  
*Operations Research student building production software for real businesses. Seeking Summer 2027 internships.*

## EDUCATION

### Cornell University, David A. Duffield College of Engineering

*Expected May 2029*

Bachelor of Science in Operations Research and Engineering | GPA: 3.82

*Relevant Coursework:* CS 2110 Data Structures & OOP · ENGRD 2700 Probability & Statistics · MATH 2940 Linear Algebra

*Activities:* Kappa Theta Pi Tech Fraternity, Cornell Blockchain, Cornell Maker Club

### Regis High School

*June 2025*

*Activities:* President of Engineering Club, States Finalist PF Debater, Computer Club, Robotics Club

## EXPERIENCE

### McCallos · mccallos.com

*May 2026 – Present*

*Chief Technology Officer*

*New York, NY*

- Sole engineer at a three-person startup building compliance and operations software for NYC property managers as a React/TypeScript web app and Capacitor iOS/Android apps on Firebase, with dedicated interfaces for doormen, superintendents, and managers
- Consolidated **17 city open-data feeds across six agencies** into a single compliance calendar covering **14 local laws and 36 obligations**, each mapped to its own penalty schedule, from **\$25-per-day** housing violations to **\$268 per ton** of excess emissions
- Built the work-order system that converts compliance obligations and maintenance requests into assigned vendor jobs, with photo-based manager approval, automated handling of routine tasks, and an evidence log recording every step

### Empire Environmental

*Brooklyn, NY*

*AI Engineer*

*June 2026 – August 2026*

- Developed an AI agent on Hermes that drafts DDC Site Safety Plan filings from any project's bid set of specifications and scanned drawings, built so the firm can absorb the retirement of its **30-year** documentation specialist without a replacement hire
- Built the agent's skills and a three-tier retrieval memory with citation write-back, then a scoring harness grading every draft against the firm's approved filings, and iterated the skills and references off the scoreboard to **98/100 across 25 templates** at **2.4 minutes** each
- Replaced text extraction with a vision pipeline for scanned drawings on a live contract's **294-page bid set**, reducing one query from **413 seconds** and **3.79M tokens** to **14 seconds** and **53K**, on a **flat \$200/month subscription** rather than metered API billing

*CAD & Design Associate*

*June 2025 – August 2025*

- Produced and revised CAD drawings across **3 active environmental remediation projects**, standardizing documentation formats to reduce design-review revision cycles

### Freelance Web Developer

*May 2026 – August 2026*

*Independent Contractor · Two paid client engagements*

*Brooklyn, NY / Red Bank, NJ*

- Designed and launched a React/TypeScript site for a two-location Italian restaurant with **1,150+ Google reviews**, replacing a defunct site and introducing the business's first online catering-inquiry system
- Increased organic clicks **140%** and impressions **134%** to **25.3K** within weeks of launch by consolidating a split domain that had been diverting **77% of traffic** and 301-redirecting **27 legacy URLs** to preserve existing rankings
- Built a transactional email pipeline on Firebase Cloud Functions that routes catering requests to staff and sends branded customer confirmations, with rate limiting, delivery logging, and recipient validation
- Delivered a React/TypeScript site for a personal styling studio with embedded Calendly consultation booking, an animated portfolio, and a design system extended from the site into print collateral

### GE Aerospace

*Bohemia, NY*

*Engineering Intern*

*August 2024*

- Wired, tested, and validated a **24V dual-battery circuit** to drive a repurposed **100W, 2500 RPM** motor, confirming **28V output** and **5.5A draw** by multimeter and calculating a **6-hour full charge cycle**
- Engineered a handheld aquatic propulsion device in Fusion 360 and OnShape, 3D printing every part and iterating on waterproofing, motor fit, and propeller safety to deliver an **18 lb working prototype** to GE program leads

## PROJECTS

### Machine Learning Stock Trading Bot

*Python, PyTorch, SQLite*

- Built an LSTM next-day price model in PyTorch trained on **12 years of daily closes** with 60-day rolling windows, supplied by a yFinance data pipeline into SQLite for backtesting
- Trained on an **80/20 train-test split** with MinMax scaling, Adam optimization, and a StepLR scheduler, rolled the model forward recursively to forecast closes several days ahead, and averaged results across **50 independent training runs**

## ADDITIONAL EXPERIENCE

### Bay Ridge Fish Bar

*October 2021 – October 2024*

*Manager*

*Brooklyn, NY*

- Managed shift staffing, cash reconciliation, and service execution for a high-volume seafood restaurant over three years
- Built an Excel inventory system tracking stock lifecycle and buy/sell timing, **reducing food waste 30% and raising margins 5%**

### Young Minds Matter

*September 2023 – June 2025*

*Volunteer Tutor*

*Manhattan, NY*

- Guided **10+ students and families** through the NYC high school application process, with **90% admitted to their top-choice school**

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, Java, JavaScript/TypeScript, SQL, React, Node.js, PyTorch, Firebase, Cloud Functions, Capacitor, Stripe, Git, Docker

**AI Engineering:** Anthropic & OpenAI APIs, retrieval pipelines, LLM evaluation harnesses · **Design:** Fusion 360, OnShape, AutoCAD